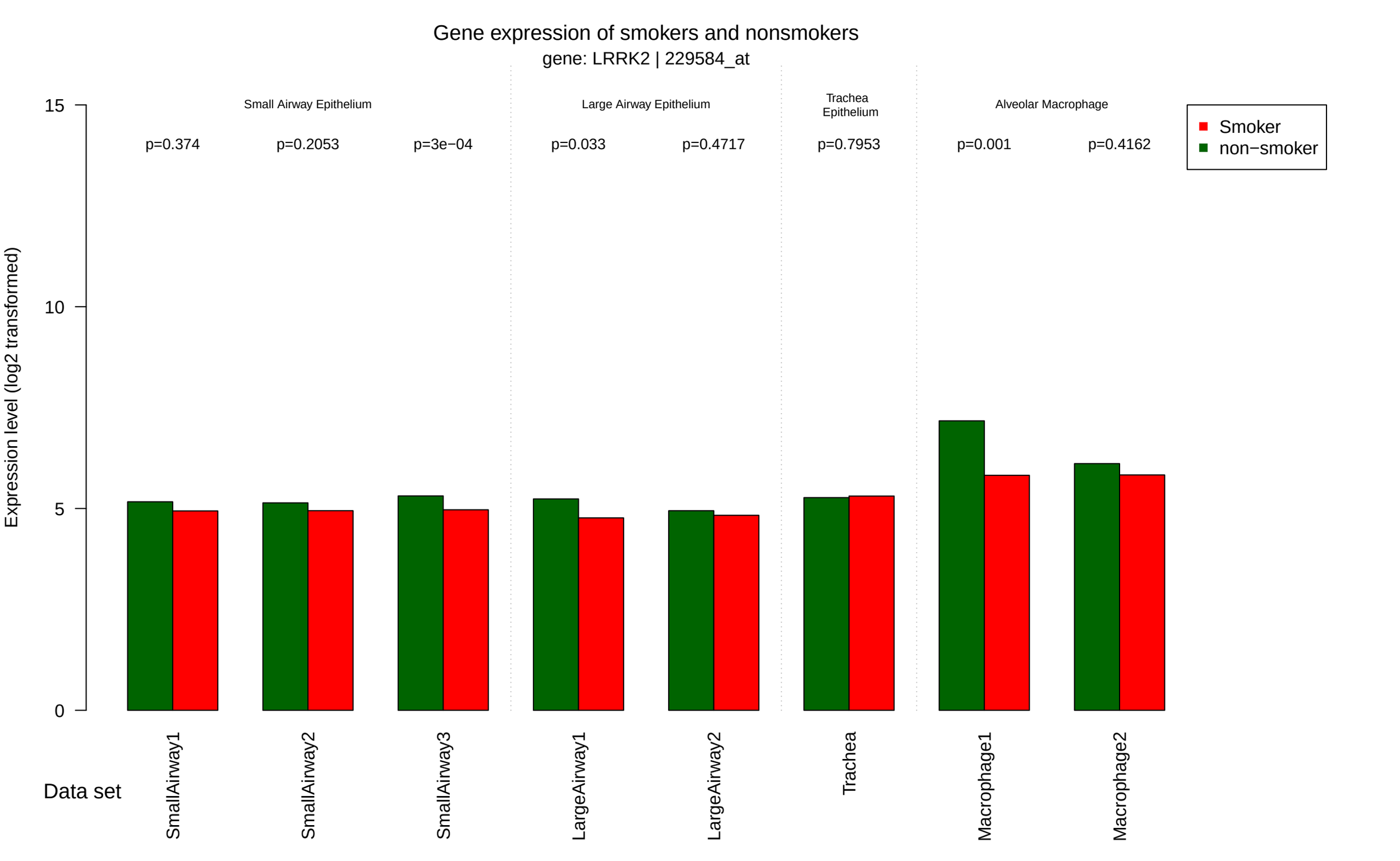
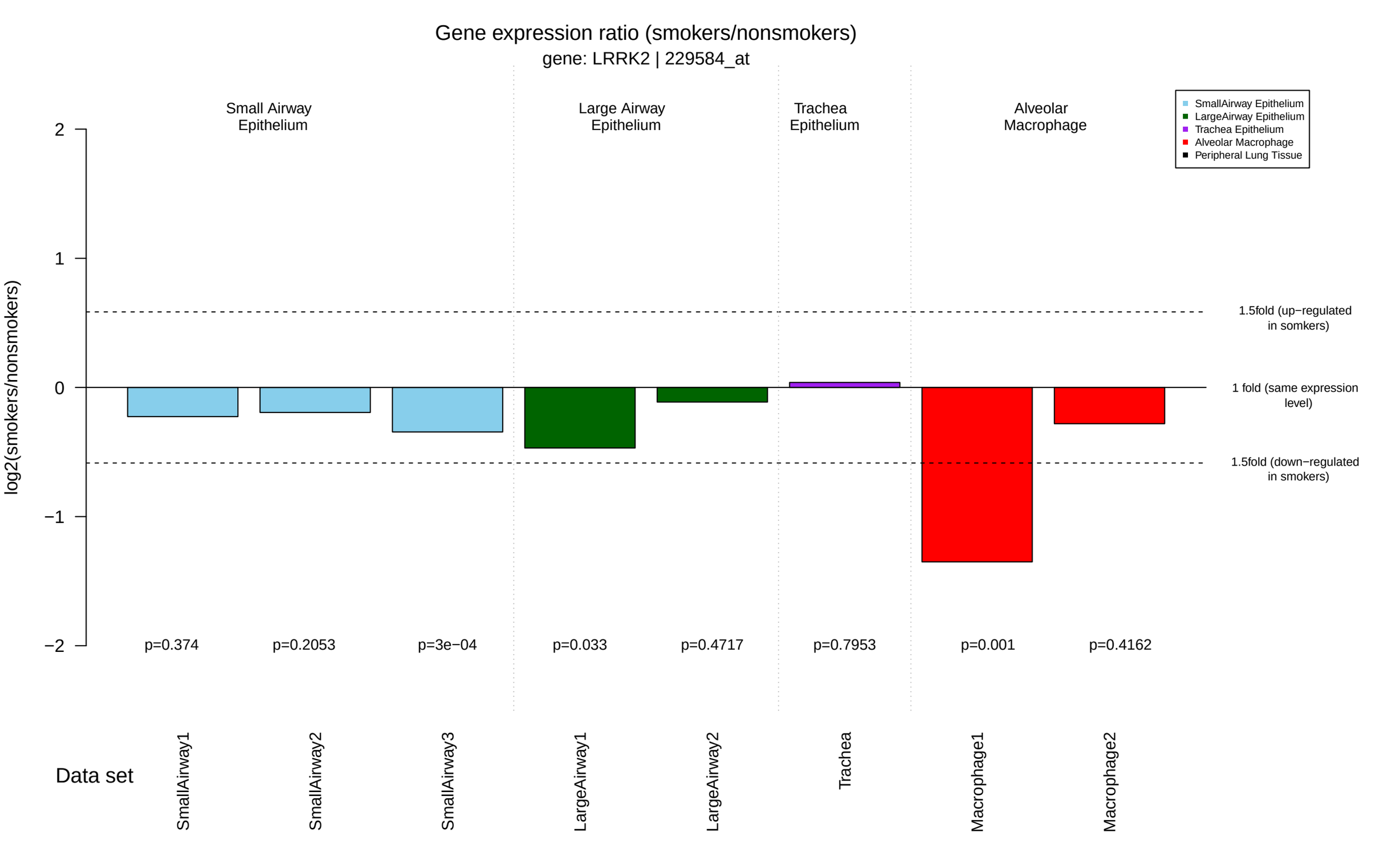
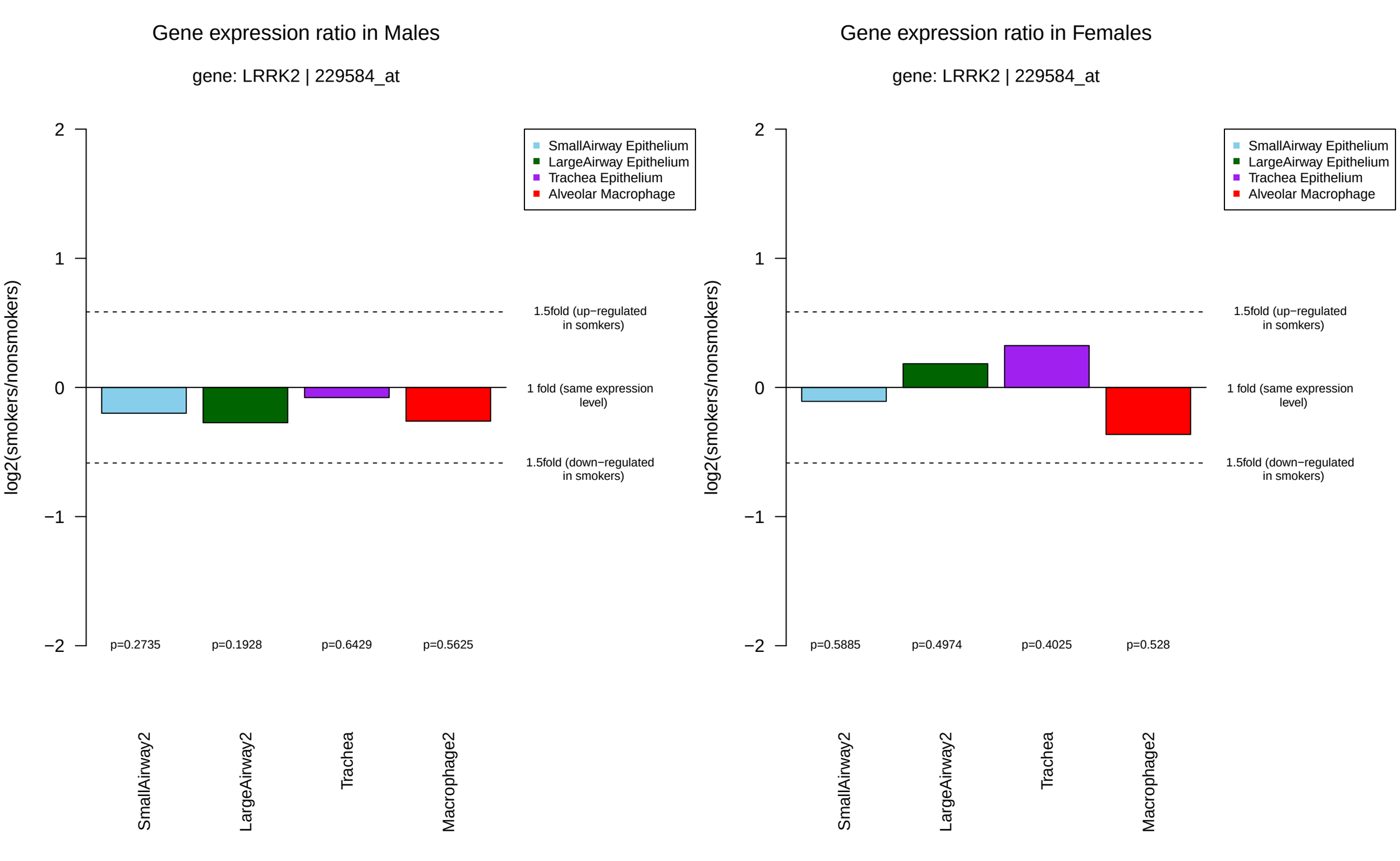
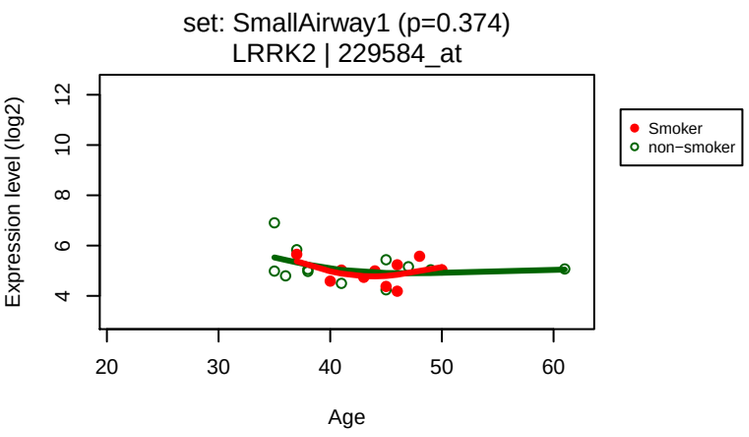
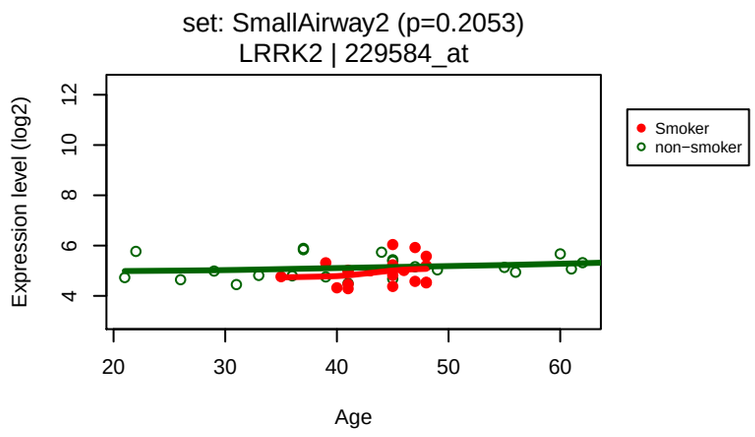
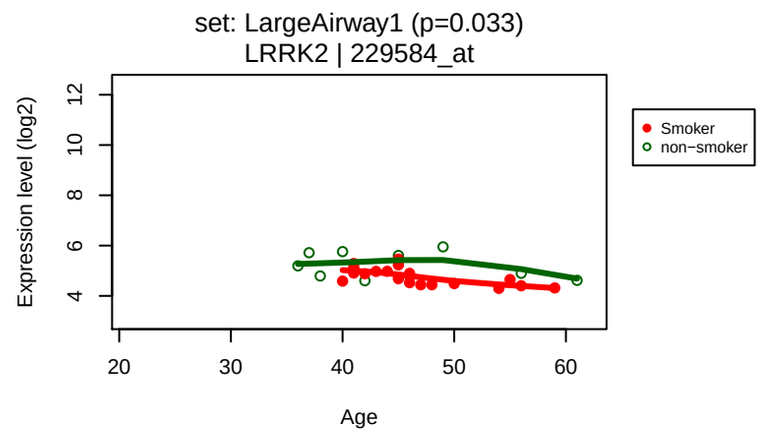

Gene expression of smokers and nonsmokers
gene: LRRK2 | 229584_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.374
p=0.2053
p=3e−04
p=0.033
p=0.4717
p=0.7953
p=0.001
p=0.4162
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

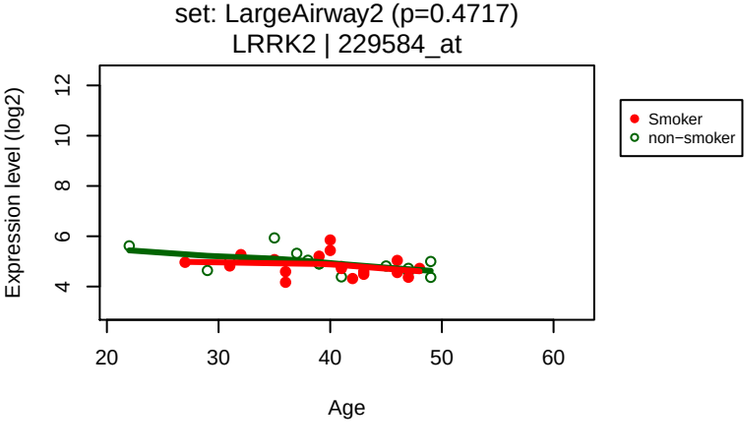

Gene expression ratio (smokers/nonsmokers)
gene: LRRK2 | 229584_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.374
p=0.2053
p=3e−04
p=0.033
p=0.4717
p=0.7953
p=0.001
p=0.4162
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

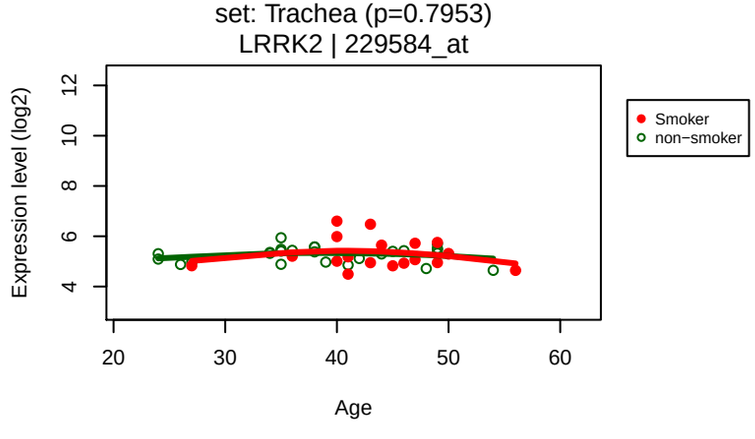

Gene expression ratio in Males
gene: LRRK2 | 229584_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2735
p=0.1928
p=0.6429
p=0.5625
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LRRK2 | 229584_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5885
p=0.4974
p=0.4025
p=0.528
SmallAirway2
LargeAirway2
Trachea
Macrophage2

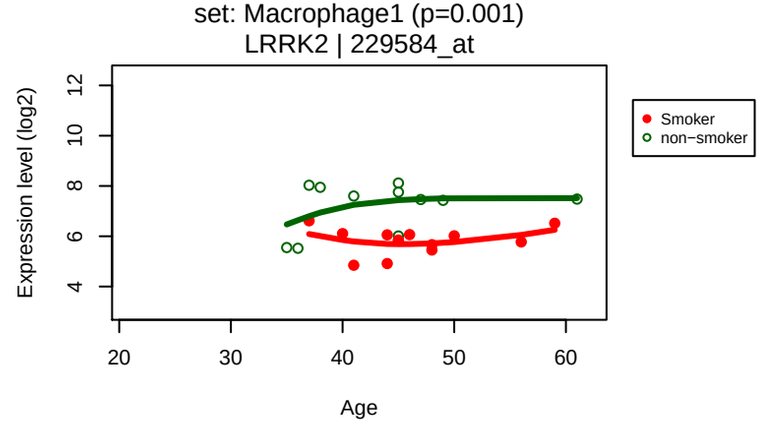

set: SmallAirway1 (p=0.374)
LRRK2 | 229584_at
Expression level (log2)
Age
Smoker
non−smoker

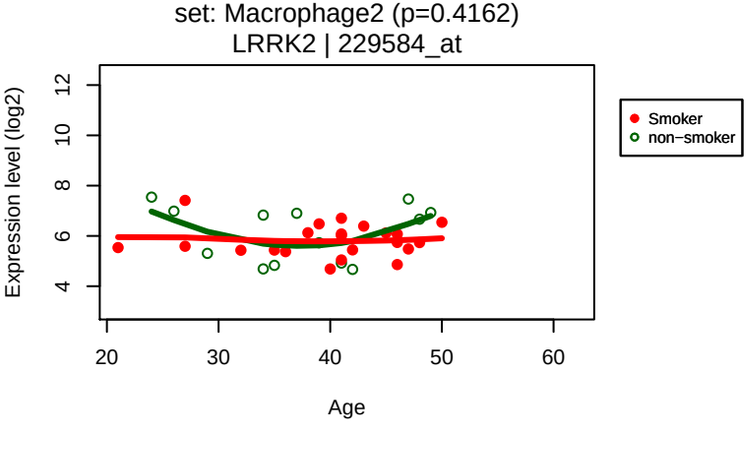

set: SmallAirway2 (p=0.2053)
LRRK2 | 229584_at
Expression level (log2)
Age
Smoker
non−smoker

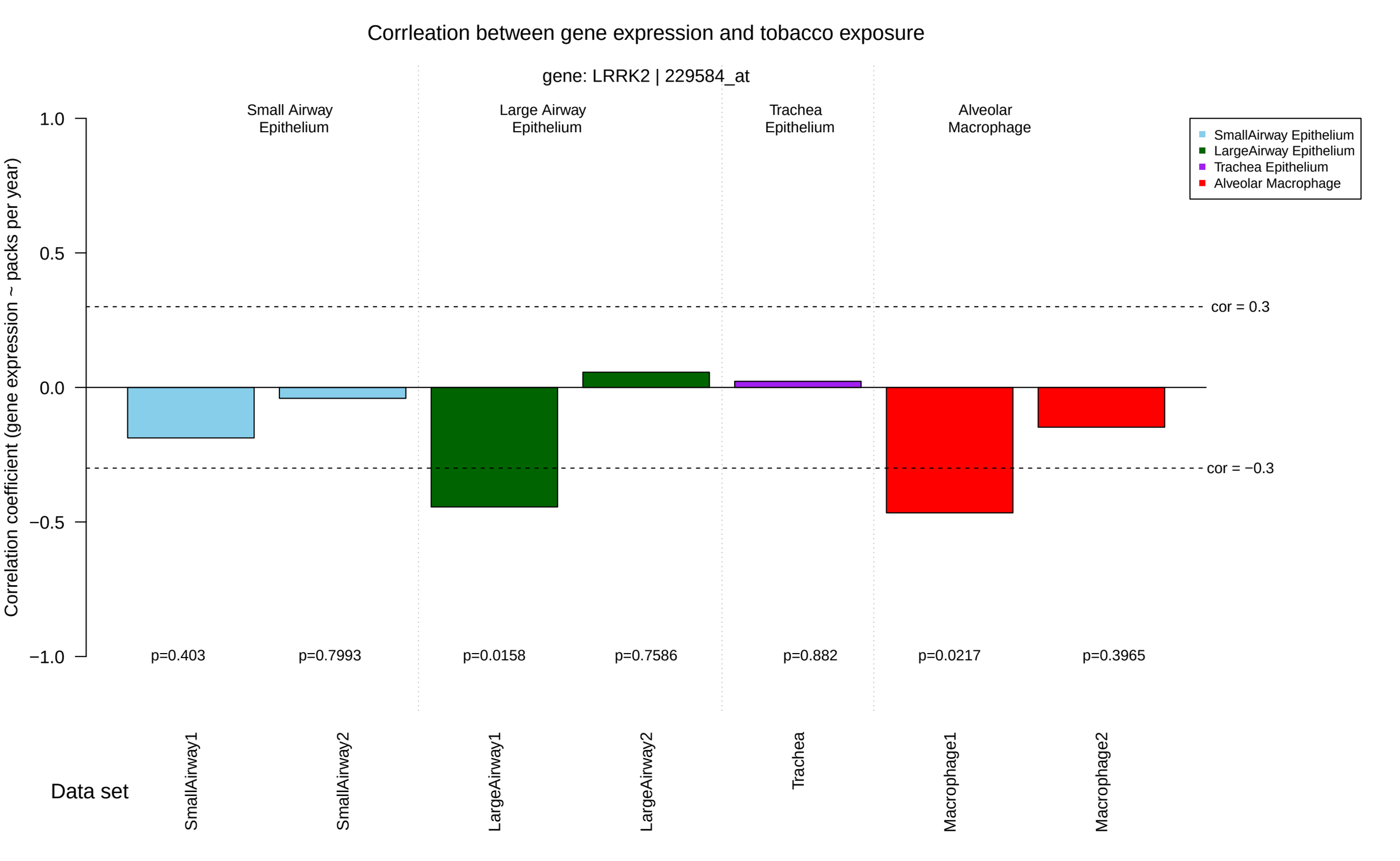

set: LargeAirway1 (p=0.033)
LRRK2 | 229584_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4717)
LRRK2 | 229584_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7953)
LRRK2 | 229584_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.001)
LRRK2 | 229584_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4162)
LRRK2 | 229584_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LRRK2 | 229584_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.403
p=0.7993
p=0.0158
p=0.7586
p=0.882
p=0.0217
p=0.3965
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2